

Test-Time Training with Masked Autoencoders: Online Version (E)

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Abstract

Generalization under distribution shifts remains a critical challenge in computer vision. Test-time training (TTT) addresses this by adapting models dynamically during deployment, using self-supervised learning to improve performance on unseen test distributions. This report evaluates the TTT-MAE framework, which integrates Masked Autoencoders into TTT, for its effectiveness on the ImageNet-C benchmark. Through experiments, we confirm TTT-MAE’s capability to enhance robustness under all corruption types. We analyze failure cases, hypothesizing a decoupling between the reconstruction and classification tasks. Additionally, an online variant of TTT-MAE, where encoder weights are not reset between test samples, demonstrates notable improvements, suggesting potential for cumulative adaptation. Despite resource constraints influencing the evaluation protocol, the findings provide critical insights into TTT-MAE’s strengths and limitations, paving the way for future refinements to optimize its performance in real-world applications. The code is available at https://github.com/killian31/ttt_mae_online

1. Introduction

Generalization under distribution shifts is a fundamental challenge in computer vision. Traditional supervised learning methods assume that training and testing data are sampled from the same distribution, a condition rarely met in real-world scenarios. As a result, models often fail to perform reliably when faced with unseen test distributions. To address this, test-time training (TTT) has emerged as a paradigm that adapts models on-the-fly during deployment, using self-supervision on individual test inputs to improve performance.

The work of Gandelsman et al. [1] shows that incorporating Masked Autoencoders (MAE) into TTT significantly improves generalization on visual benchmarks with distribution shifts.

This report evaluates the effectiveness of TTT-MAE and extends the analysis in several directions. We investigate the method’s performance on ImageNet-C [2], analyze fail-

ure cases, and implement an online version of TTT-MAE, where the encoder’s weights are not reset after each test image.

2. Data

ImageNet-C [2] is a benchmark designed to evaluate model robustness under distribution shifts. It consists of the ImageNet validation set with 15 types of corruptions applied, each with five levels of severity. These corruptions introduce controlled distribution shifts, making ImageNet-C a suitable benchmark for studying test-time adaptation techniques.

For our experiments, we use the same data augmentation strategy as in TTT-MAE [1], which involves applying random crops and horizontal flips during training. Figure 1 illustrates examples of different corruption types from ImageNet-C, showcasing the diverse challenges posed by these perturbations.

3. Model Architecture

The model architecture follows the design proposed in [1], consisting of a Y-shaped structure with a shared encoder and two separate heads. The encoder is a ViT-Large model, pre-trained as a MAE. The first head is a lightweight decoder used for the self-supervised reconstruction task, while the second head is a classifier.

4. Test-Time Training

At test time, the encoder and self-supervised head are adapted for each test input using a reconstruction loss. Specifically, 75% of the input image patches are randomly masked, and the model optimizes the mean squared error between the reconstructed and original patches. This adaptation enables the model to adjust dynamically to distribution shifts without requiring prior knowledge of the test distribution. The classification head remains fixed during TTT, ensuring that the learned features are effectively transferred to the main task. To evaluate the method’s robustness, predictions are made after adapting the encoder and self-supervised head to each test input. Each test input is treated independently, and the model weights are reset to

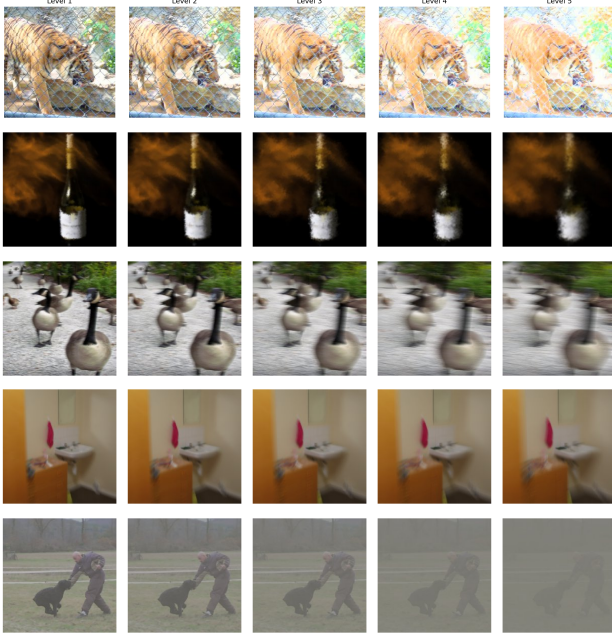


Figure 1. Example of corruption types Brightness, Glass Blur, Motion Blur, Zoom Blur, Contrast (from top to bottom) for the 5 levels of corruption.

their pre-trained state before processing the next input.

5. Experiments

5.1. Paper’s results

While TTT-MAE achieves significant improvements under distribution shifts, it is computationally expensive to evaluate. The model, based on a ViT-Large encoder, has 416,305,384 parameters.

Additionally, the method processes test samples independently, preventing batch processing of test images and further increasing the computational burden.

Given these constraints, I made modifications to the evaluation protocol to reduce the computational requirements:

1. The original protocol evaluates 50 images per category for each corruption type across all 1,000 classes, amounting to 50,000 test images per corruption type. Due to resource limitations, I evaluated only 2 images per category, resulting in 2,000 test images per corruption type and $15 \times 2,000$ in total across all corruptions.
2. The batch size used to compute gradients during TTT was reduced from 128, as in the paper, to 16. A smaller batch size may compromise the quality of gradient approximation and potentially affect the adaptation quality.

These adjustments are necessary to manage the computational demands but can also explain discrepancies between my results and those reported in the paper. For in-

stance, while the original method consistently outperforms the baseline across all corruption types, my results show four corruption types (namely Brightness, Defocus blur, Fog and Frost) where TTT fails to surpass the baseline. Conversely, my accuracy for Gaussian noise is significantly higher than reported in the paper. These inconsistencies highlight the trade-offs made to accommodate resource constraints. Despite these limitations, the evaluation still provides valuable insights into the TTT-MAE method and its applicability under resource-constrained scenarios. Figure 2 shows my full results compared to the original baseline and results of TTT-MAE.

Figure 3 provides more details on the evolution of accuracy along TTT steps.

Looking at the four corruptions which failed, they all display a drop in accuracy between TTT step 1 and step 2, suggesting the model might have stuck in some local minimum at the beginning of step 1, maybe because of a too large learning rate. Results from the baselines in section 5.3, table 1 confirm this, since they are more similar to the results of [1], and were obtained with a much lower learning rate of 3.13×10^{-4} instead of 5×10^{-3} , more adapted to the smaller batch size.

It looks like increasing the number of steps results in increasing the accuracy, as speculated in [1], except for the four corruption types which did not beat the baseline. In section 5.2, we will try to characterize what is going on during TTT for Defocus blur, visualizing some samples and evolution of metrics.

5.2. Failure cases

For this experiment, I chose to analyze two samples from the same category (*tench*, *Tinca tinca*), one that the models fails to classify and one that the models classifies correctly, for the *Defocus blur* corruption type, as it is the corruption type with the lowest accuracy. I run TTT for 100 steps on the two images from, and log the reconstruction and classification losses at each iteration. The results are displayed in figure 4. The first image is hardly recognizable even for a human, and the second one is easier to identify as a fish.

The classification loss during TTT for the first sample (top-row of figure 4) shows a chaotic behavior, looking random, and even starts increasing after step 60, ending at a greater value than at step 0. This image is consistently misclassified at each TTT step. Notice that the reconstruction loss is low at each TTT step (close to 0), and remains stable across all steps, not increasing nor decreasing, indicating a decoupling between the reconstruction task and downstream classification.

For the second sample (bottom-row of figure 4), the recon-

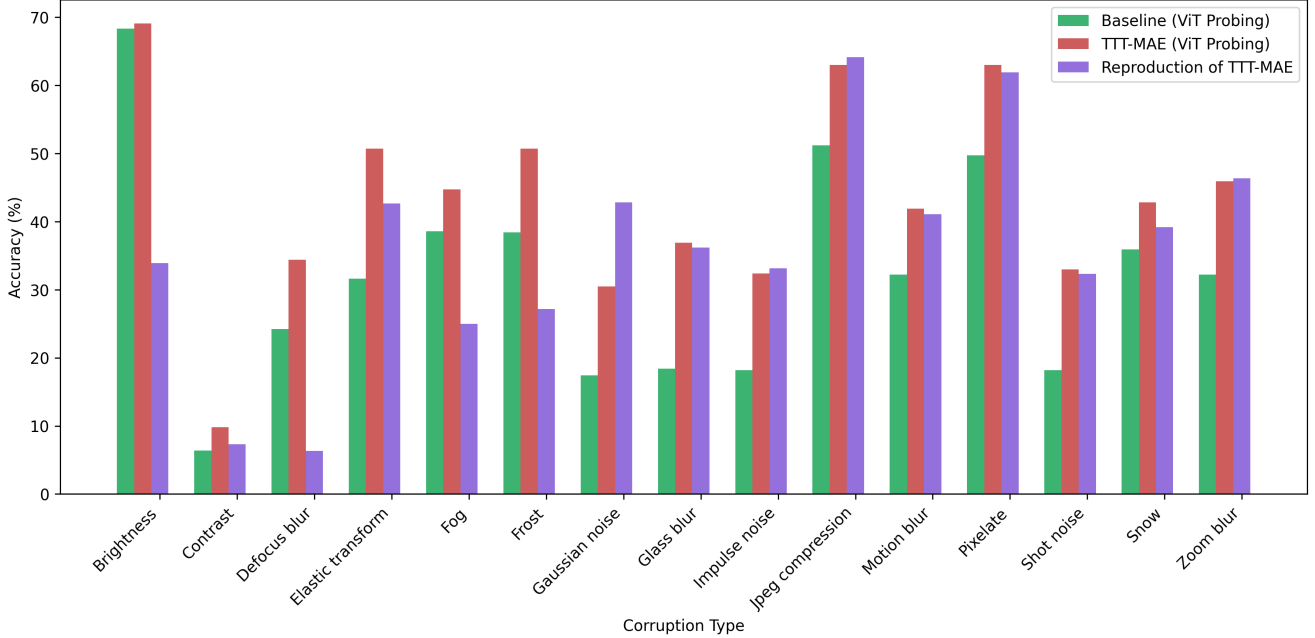


Figure 2. Accuracy on ImageNet-C, level 5.

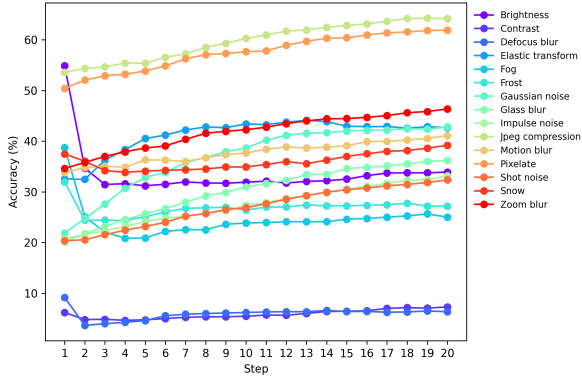


Figure 3. Accuracy vs. number of steps for all corruption types with TTT-MAE on ImageNet-C, level 5.

struction loss is similar to the one of the first sample, stable and close to 0, while the classification loss has a different behavior, reinforcing the hypothesis of a reconstruction-classification decoupling.

This sample is correctly classified at almost every TTT step, except for two spikes in the classification loss before step 33. After step 33, the classification loss is extremely low and remains stable until step 100.

This highlights the importance of the number of TTT-steps. Setting 30 steps here would have resulted in incorrectly classify the second image, whereas it was well classified after the first step.

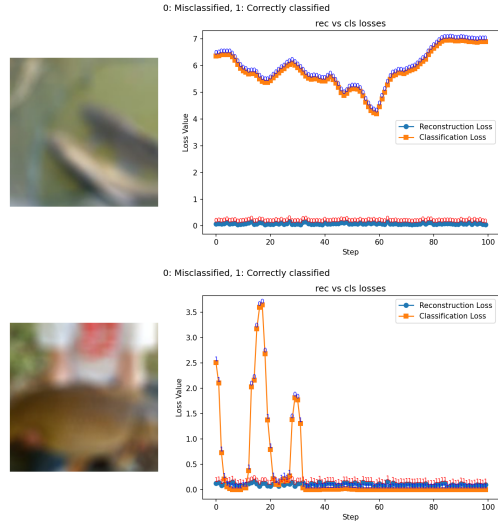


Figure 4. Reconstruction loss, classification loss and prediction at each TTT step, for two samples of the *Defocus blur* corruption type, category *tench*, *Tinca tinca*.

Looking at the reconstructed image from the decoder (figure 5) does not suggest that the models' reconstruction ability improves, following the stability of the reconstruction loss.

5.3. Online TTT-MAE

For this last experiment, we will investigate an online version of TTT, where the model's weights are not reinitialized

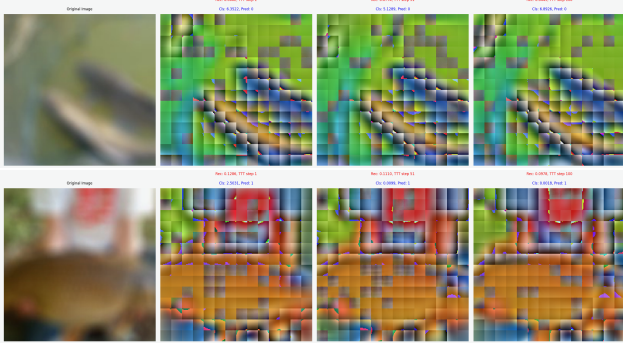


Figure 5. Reconstructed samples at different TTT steps, for two samples of the *Defocus blur* corruption type, category *tench*, *Tinca tinca*.

Corruption	Baseline TTT	TTT Online
Brightness	69.5	74.1
Defocus blur	26.55	34.15
Frost	43.7	53.95

Table 1. TTT-MAE Online and baseline accuracy results (%) on ImageNet-C corruption types Brightness, Defocus blur and Frost, level 5.

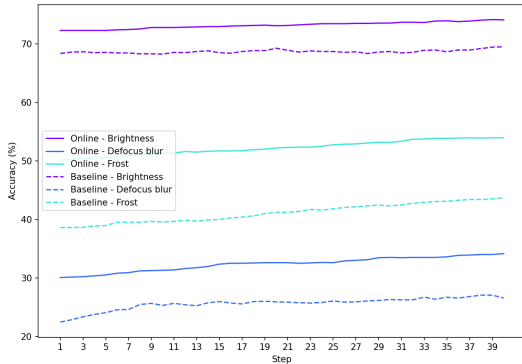


Figure 6. TTT-MAE Online: accuracy vs TTT step.

between each test sample. For this, I ran the algorithm on three corruption types (Brightness, Defocus blur and Frost), with a batch size of 1 and two samples per ImageNet category. I used SGD as optimizer, and ran a TTT baseline with this setting for each of the three corruption types. Results are reported in table 1.

We can observe a significant improvement in accuracy with the online version. As shown in figure 6, the improvement occurs even at a low number of TTT steps.

6. Discussion

This report evaluates the effectiveness of TTT-MAE on the ImageNet-C dataset, providing insights into its performance and areas for potential improvement. The results demonstrate that TTT-MAE generally enhances robustness under distribution shifts, as observed in all corruption types. The analysis of the failure cases suggests a potential decoupling between the reconstruction task and downstream classification, though this hypothesis requires further investigation to confirm.

Adjustments to the evaluation protocol, necessitated by resource constraints, may have contributed to discrepancies between my findings and the original results reported by [1]. Nonetheless, these modifications do not diminish the utility of the insights gained, especially regarding the stability of TTT-MAE’s adaptation process and its sensitivity to hyperparameters such as the number of TTT steps. An important finding is the high sensitivity of TTT-MAE to the learning rate value, which can significantly degrade the performance if not carefully chosen.

The exploration of an online variant of TTT-MAE reveals substantial improvements in performance, particularly for corruption types where the baseline struggles. This approach suggests that cumulative updates across test samples can enhance adaptation without compromising efficiency. However, further analysis is needed to assess the long-term stability and broader applicability of this online approach.

Overall, TTT-MAE proves to be a powerful method for addressing distribution shifts, with room for refinement. Future work should focus on deepening the understanding of failure cases, optimizing the alignment of self-supervised and classification tasks by experimenting a different reconstruction loss, and validating the method’s robustness across diverse datasets.

References

- [1] Yossi Gandelsman, Yu Sun, Xinlei Chen, and Alexei A Efros. Test-time training with masked autoencoders. In *Advances in Neural Information Processing Systems*, 2022. 1, 2, 4
- [2] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. *Proceedings of the International Conference on Learning Representations*, 2019. 1